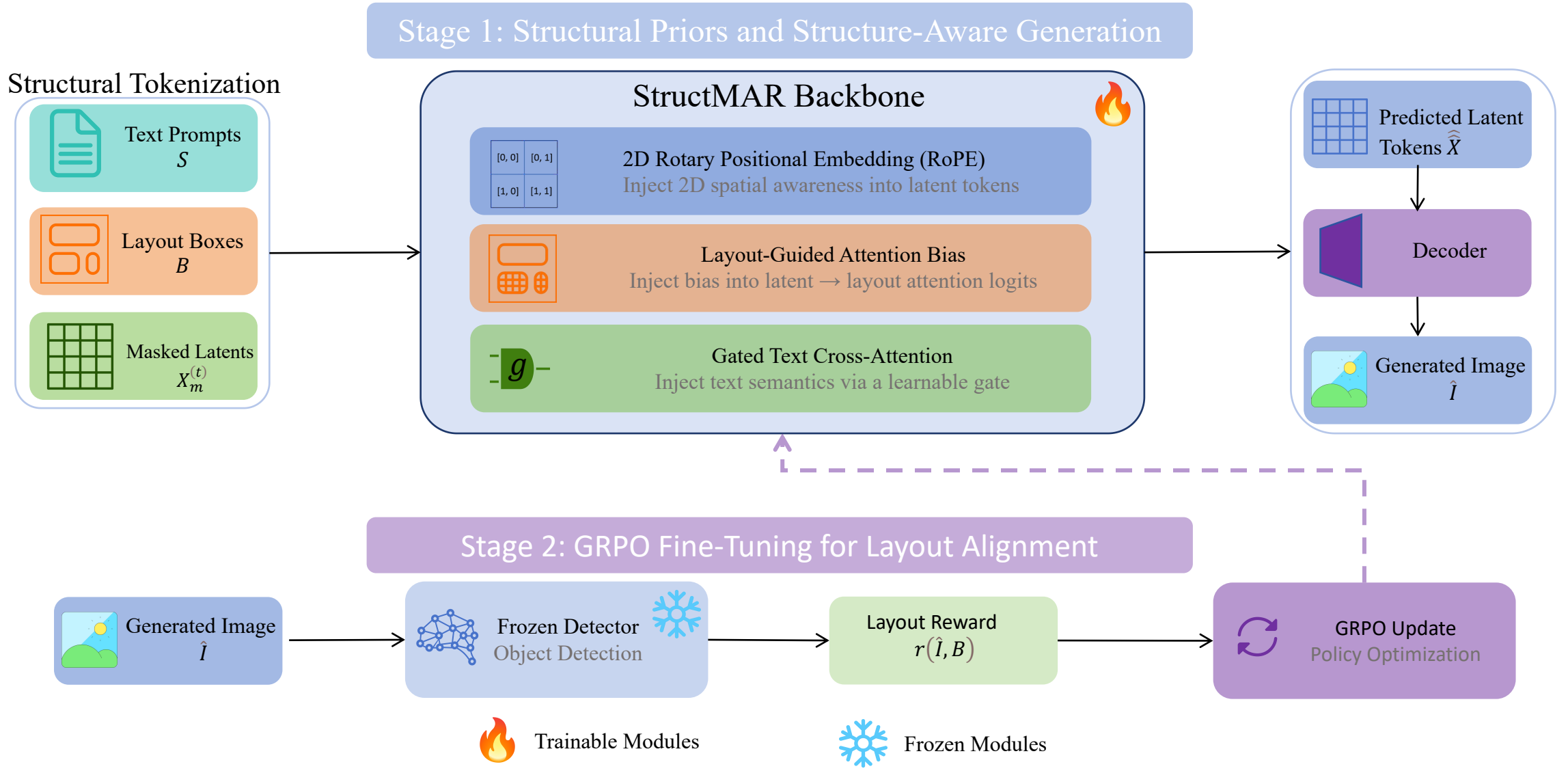
Stage 1: Structural Priors and Structure-Aware Generation
Structural Tokenization
Text Prompts
S
Layout Boxes
B
Masked Latents
$X_m^{(t)}$
StructMAR Backbone
[0, 0]
[0, 1]
[1, 0]
[1, 1]
2D Rotary Positional Embedding (RoPE)
Inject 2D spatial awareness into latent tokens
Layout-Guided Attention Bias
Inject bias into latent → layout attention logits
Gated Text Cross-Attention
Inject text semantics via a learnable gate
Predicted Latent Tokens $\hat{X}$
Decoder
Generated Image
$\hat{I}$
Stage 2: GRPO Fine-Tuning for Layout Alignment
Generated Image
$\hat{I}$
Frozen Detector
Object Detection
Layout Reward
$r(\hat{I}, B)$
GRPO Update
Policy Optimization
Trainable Modules
Frozen Modules